MAXIMAL EVENNESS VIA DISCREPANCY THEORY

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Abstract. A widely studied property in mathematics, physics, and music theory is that of *maximal evenness*, in which a fixed number of points are distributed "as evenly as possible" in $\mathbb{Z}/k\mathbb{Z}$. There exist many equivalent definitions of maximal evenness, and here we propose and explore a new definition based on discrepancy theory. First, we prove that this definition is equivalent to the original definition given by Clough and Myerson. Then, we use this characterization to develop a fast algorithm (linear in time and space) for checking whether a given set is maximally even. Finally, we use the discrepancy characterization to define a notion of maximal evenness for the spaces $\mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_d\mathbb{Z}$ and $\mathbb{Z}$.

1. Introduction

Consider the following combinatorial problem: Given fixed integers k and n, which sets $A \subseteq \mathbb{Z}/k\mathbb{Z}$ with $|A| = n$ achieve the "most even" distribution? Such sets are referred to as *maximally even* sets, and they are central objects of study in mathematics, physics, and music theory [7].

The concept of maximal evenness was first introduced by Clough and Myerson in [6] in the context of tonal and atonal music theory, in which elements of $\mathbb{Z}/k\mathbb{Z}$ are referred to as *pitch classes* (or *pc*s) and subsets of $\mathbb{Z}/k\mathbb{Z}$ are referred to as *pitch class sets* (or *pc set*s). In the subsequent decades, many music theorists and mathematicians began to formulate different precise notions of "even distribution", many of which turn out to be equivalent to the definition of Clough and Myerson; see Section 4, Theorem 7 of [7] for an outline of a few of these equivalent definitions.

These equivalent definitions of maximal evenness are motivated by a variety of mathematical disciplines: number theory [5], combinatorial optimization [3], discrete Fourier analysis [2], and, in our case, discrepancy theory. In accordance with these many connections accross mathematics, these ideas first came to me in the context of applied topology [1]: While studying the homotopy types and persistent homology of the regular polygons, it was observed that the edges of P_n which are touched by an inscribed star must form (what I now know to be called) a maximally even set.

In this paper, we contribute another equivalent definition to this list. Our approach is based on *discrepancy theory*, as developed by Erdős and Turán in [9]. The language of discrepancy theory is very appealing when discussing maximal evenness, as it provides a useful means of quantifying how much a set of points fails to be equidstirbuted in the circle. Discrepancy theory has found a wide range of applications across mathematics and computer science, including number theory [8], quasi-Monte-Carlo integration [13], and computational geometry [4].

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Our main contributions are as follows: First, we prove that a set $A \subseteq \mathbb{Z}/k\mathbb{Z}$ with $|A| = n$ is maximally even if and only if its discrepancy is bounded above by $1/n+1/k$. Then we use this characterization to construct a fast algorithm (linear in time and in space) which checks whether a given set is maximally even. Finally, we use the discrepancy characterization to generalize the notion of maximal evenness to subsets of spaces more general than $\mathbb{Z}/k\mathbb{Z}$.

2. Definition

Consider any set $A \subseteq \mathbb{Z}/k\mathbb{Z}$ with $|A| = n$. Let the tuple (k, n) be called the *type* of the set A. Now, it will prove useful to be able to reimagine any such A in a variety of different ways that make certain properties more or less salient. First, we may think of a subset A as a *binary string* by considering the indicator function $\mathbf{1}_A : \mathbb{Z}/k\mathbb{Z} \to \{0, 1\}$. Towards this end, we may write such a string explicitly as $0^{g_0}10^{g_1}10^{g_2}\dots 10^{g_n}$, where $g_0, g_1, \dots g_n$ are nonnegative integers with $n + \sum_{e=0}^{n} g_e = k$. Alternatively, we may think of A as the set $\{\omega_k^i : i \in A\}$ (where ω_k is the kth root of unity) so that we may study the geometry of A within the unit circle $\mathbb{S}^1$. We will use these three interpretations interchangeably throughout the paper; it will usually be clear from context which meaning is intended.

Next, we invest in some new notation that will be useful for the remainder of this paper. For indices $i, j \in \mathbb{Z}/k\mathbb{Z}$, let $\vec{d}_k(i, j)$ represent the *oriented distance* from i to j in $\mathbb{Z}/k\mathbb{Z}$. That is, $\vec{d}_k(i, j) = j - i$ if $i \leq j$, and $\vec{d}_k(i, j) = k - j + i$ if $i > j$. Moreover, we define $[i, j)_k$ to be the interval $[\omega_k^i, \omega_k^j) \subseteq \mathbb{S}^1$ and likewise for other types of intervals. We also let λ be the Lebesgue (probability) measure on $\mathbb{S}^1$. Notably, these definitions give $\vec{d}_k(i, j)/k = \lambda([i, j)_k)$.

This in mind, we can recall the definition of maximal evenness. While the following definition is not the exact language used by Clough and Myerson in [6], it is easily seen to be equivalent:

Definition 2.1. For any set A of type (k, n), we say that A is *maximally even* if any elements $i, j \in A$ give

$$\left| \vec{d}_k(i, j) - \frac{k}{n} \#(A \cap [i, j)_k) \right| < 1. \tag{1}$$

What Clough and Myerson identified is that the values $\vec{d}_k(i, j)$ ("length") and $A \cap [i, j)_k$ ("mass") should be dual to each other, in the sense that the value of one should almost determine the value of the other. In this way, it is clear that a set A which fails to be maximally even must have some interval whose length and mass are not in accordance. But observe that $\vec{d}_k(i, j)$ is "too long" compared to $A \cap [i, j)_k$ if and only if $\vec{d}_k(j, i)$ is "too short" compared to $A \cap [j, i)_k$. Using this fact, we can reduce the amoung of case-checking that may be required when working with sets that are known to not be maximally even.

Likewise, let us recall the definition of discrepancy given in [9]:

Definition 2.2. For X a finite set of points in $\mathbb{S}^1$, the *discrepancy* of X is

$$D(X) = \sup_{I \subseteq \mathbb{S}^1} \left| \frac{\#(X \cap I)}{\#X} - \lambda(I) \right| \tag{2}$$

where the supremum is taken over all intervals (connected subsets) of $\mathbb{S}^1$.

We observe that there seems to be some relation between (1) and (2): The value $\lambda(I)$ corresponds to $\vec{d}_k(i,j)/k$ and the value $\#(A \cap [i,j)_k)/n$ corresponds to $\#(X \cap I)/n$. Hence, it appears we may be able to cast maximal evenness in terms of discrepancy by making this correspondence precise. Indeed, we do this by the following lemma:

Lemma 2.3. *A set A of type (k,n) is maximally even if and only if and any indices $i,j \in \mathbb{Z}/k\mathbb{Z}$ give*

$$(*) \qquad \left| \#(A \cap [i,j)_k) - n\frac{\vec{d}_k(i,j)}{k} \right| < 1.$$

Proof. First suppose that A of type (k,n) is maximally even, and let $i,j \in \mathbb{Z}/k\mathbb{Z}$ be arbitrary indices. We may then choose indices $i',j' \in A$ such that $\#(A \cap [i,j)_k) = \#(A \cap [i',j')_k)$ and $|\vec{d}_k(i',j') - \vec{d}_k(i,j)| \leq \frac{k}{n} - 1$ both hold. Then we can bound:

$$\begin{aligned} \left| \#(A \cap [i,j)_k) - \frac{n}{k}\vec{d}_k(i,j) \right| &\leq \frac{n}{k}\left| \vec{d}_k(i',j') - \frac{k}{n}\#(A \cap [i',j')_k) \right| + \frac{n}{k}\left| \vec{d}_k(i',j') - \vec{d}_k(i,j) \right| \\ &< \frac{n}{k} + \frac{n}{k}\left(\frac{k}{n} - 1\right) = 1. \end{aligned}$$

Conversely, suppose that A is not maximally even. That is, there are indices $i,j \in A$, such that we have

$$\vec{d}_k(i,j) \geq \frac{k}{n}\#(A \cap [i,j)_k) + 1. \tag{3}$$

In particular, let us assume that i,j are such that they minimize the value $\vec{d}_k(i,j)$ over all pairs $i,j \in A$ satisfying (3). Then the interval $[i+1,j)_k$ must satisfy $\#(A \cap [i+1,j)_k) = \#(A \cap [i,j)_k) - 1$ and $\vec{d}_k(i+1,j) = \vec{d}_k(i,j) - 1$. Plugging this into (3) and rearranging gives

$$\#(A \cap [i+1,j)_k) \leq \frac{n}{k}\vec{d}_k(i+1,j) - 1. \tag{4}$$

Hence, the indices $i+1, j \in \mathbb{Z}/k\mathbb{Z}$ violate $(*)$. □

Finally, we can state our first important result.

Theorem 2.4. *A set A of type (k,n) is maximally even if and only if it satisfies the discrepancy bound $D(A) < 1/n + 1/k$.*

Proof. First, let A be a set of type (k,n), and suppose that A is maximally even. Then note that, for any inteval $I \subseteq \mathbb{S}^1$, we may choose indices $i,j \in \mathbb{Z}/k\mathbb{Z}$ such that we get $\#(A \cap I) = \#(A \cap [i,j))$ and $|\vec{d}_k(i,j)/k - \lambda(I)| < 1/k$. Then it follows that we get

$$\begin{aligned} \left| \frac{\#(A \cap I)}{n} - \lambda(I) \right| &\leq \frac{1}{n}\left| \#(A \cap I) - n\frac{\vec{d}_k(i,j)}{k} \right| + \left| \frac{\vec{d}_k(i,j)}{k} - \lambda(I) \right| \\ &< \frac{1}{n} + \frac{1}{k}, \end{aligned} \tag{5}$$

where the second inequality follows from Lemma 2.3. But note that the transformation $I \mapsto [i,j)_k$ maps the set of all intervals $I \subseteq \mathbb{S}^1$ onto the set of all pairs of indices in $\mathbb{Z}/k\mathbb{Z}$, which is a finite set. Hence, the supremum in Definition 2.2 is taken over a finite set of values which are all strictly less than $1/n + 1/k$. The resulting value must then be strictly less than $1/n + 1/k$, so we get $D(A) < 1/n + 1/k$.

Conversely, suppose that A is not maximally even. Then by Lemma 2.3, there exist indices $i, j \in \mathbb{Z}/k\mathbb{Z}$ such that $(*)$ fails. In particular, we have

$$\#(A \cap [i,j)_k) - n\frac{\vec{d}_k(i,j)}{k} \geq 1. \tag{6}$$

Then we consider the interval $I = [i, j-1]_k$. Observe that $\#(A \cap I) = \#(A \cap [i,j)_k)$ and $\lambda(I) = \vec{d}_k(i,j)/k - 1/k$. Plugging this in to $(*)$ and rearranging, we get

$$\frac{\#(A \cap I)}{n} - \lambda(I) \geq \frac{1}{n} + \frac{1}{k}. \tag{7}$$

Hence, A fails the discrepancy bound. □

We now explore the prospect of letting Theorem 2.4 serve as the definition of maximal evenness. Indeed, it immediately makes clear that maximal evenness coincides with an asymptotic notion of equidistribution, at least in the following sense: Let $\{(k_i, n_i)\}_{i=1}^{\infty}$ be a sequence of types, let A_i be a maximally even set of type (k_i, n_i), and let $I \subseteq \mathbb{S}^1$ be some fixed interval. If it is assumed that $n_i \to \infty$ (hence $k_i \to \infty$) as $i \to \infty$, then we get $\#(A_i \cap I)/n_i \to \lambda(I)$ as $i \to \infty$. That is, the proportion of $\{A_i\}_{i=1}^{\infty}$ contained in I is determined only by the size of I. Moreover, it is immediate that A of type (k,n) is maximally even if and only if its bitwise complement $A \oplus 1^k$ is maximally even.

Additionally, it is very straightforward to generalize Theorem 2.4 to spaces other than $\mathbb{Z}/k\mathbb{Z}$. In Section 4, we use a discrepancy-based characterization to define what it means to be maximally even in $\mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_d\mathbb{Z}$ and in $\mathbb{Z}$.

With this definition, we may interpret Lemma 2.3 as a combinatorial version of Theorem 2.4 which is more amenable for use in computational and algorithmic settings.

We conclude this section by considering the following notion of equivalence:

Definition 2.5. Suppose that A_1 and A_2 are two binary strings of type (k,n). We say that A_1 and A_2 are *translation-equivalent*, denoted $A_1 \sim A_2$, if there exists some integer $r \in \mathbb{Z}/k\mathbb{Z}$ such that $r + A_1 = \{r + a : a \in A_1\} = A_2$. A *binary necklace* or *necklace* $\tilde{A}$ is an equivalence class under $\sim$, and a set A in this equivalence class is called an *unfolding* of $\tilde{A}$.

Figure 1 contains examples of binary necklaces of various types. This quotient structure is instrumental for the results of this paper, as many important properties hold for all binary strings within an equivalence class. In particular, type and maximal evenness are both translation-invariant properties. In this way, we will freely interchange between saying that a binary string has a property and that the associated binary necklace has that property.

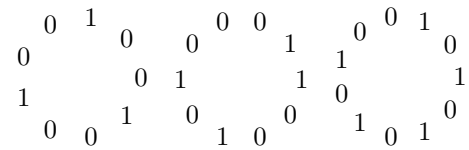

FIGURE 1. Examples of binary necklaces, with types (9,3) (left), (10,4) (center), and (11,5) (right).

3. Algorithm

As an application of Lemma 2.3/Theorem 2.4, we create a fast algorithm for checking whether a given set $A \subseteq \mathbb{Z}/k\mathbb{Z}$ is maximally even. To do this, we first consider the procedure which converts A from its representation as a set to its representation as a binary string; it is clear that such an algorithm runs in $\Theta(k)$ time and $\Theta(k)$ space. For the remainder of this section, it will be assumed that A has been stored as a binary string.

Next, we develop some machinery that will help us define the algorithm that will check if a given binary necklace is maximally even or not.

Definition 3.1. We say that a binary necklace $\tilde{A}$ of type (k, n) with $n > 0$ has *balanced gaps* if any unfolding A of $\tilde{A}$ of the form $A = 10^{g_1}10^{g_2}\dots 10^{g_n}$ has, for all $e \in \{1, \dots n\}$, either $g_e = \lfloor \frac{k}{n} - 1 \rfloor$ or $\lceil \frac{k}{n} - 1 \rceil$.

We now seek an algorithm which checks whether a given binary string has balanced gaps. Evidently, we can compute and store the size of the gaps in a binary string using a method GetGaps that runs in $\Theta(k)$ time and $\Theta(n)$ space. Using this method as a black box, we can define Algorithm 1, which evidently runs in $\Theta(k) + O(n) = \Theta(k)$ time and $\Theta(n)$ space. We also note the following property of balanced gap necklaces: If A has balanced gaps, then the number of e with $g_e = \lfloor \frac{k}{n} - 1 \rfloor$ is $n - (k \mod n)$ and the number of e with $g_e = \lceil \frac{k}{n} - 1 \rceil$ is $k \mod n$.

Algorithm 1 Checking if a binary string has balanced gaps

```
procedure HasBalancedGaps(A, k, n)
    G ← GetGaps(A)
    for e ∈ {0, ... n − 1} do
        if G[e] ≠ ⌊k/n − 1⌋ and G[e] ≠ ⌊k/n − 1⌋ then return false
    return true
```

Next, we prove the following basic properties of balanced gap necklaces.

Lemma 3.2. *If n divides k, then there is only one binary necklace with balanced gaps, and it is maximally even.*

Proof. If n divides k, then $\lfloor \frac{k}{n} - 1 \rfloor = \lceil \frac{k}{n} - 1 \rceil = \frac{k}{n} - 1$. Hence, the only possible necklace with bounded gaps is the $\tilde{A}$ which corresponds to the unfolding $A = (10^{\frac{k}{n}-1})^n$. Now take any indices $i, j \in \mathbb{Z}/k\mathbb{Z}$ and define $a = \#(A \cap [i, j))$. Then let us write $A \cap [i, j) = 0^{g_0}10^{g_1}\dots 10^{g_a}$ so that we get $g_e = \frac{k}{n} - 1$ for all $e \in \{1, \dots a-1\}$ and $0 \le g_e \le \frac{k}{n} - 1$ for $e = 0, a$. Then we may bound:

$$\left|a - \frac{n}{k}\vec{d}_k(i,j)\right| = \left|a - \frac{n}{k}(a + \sum_{e=0}^{a} g_e)\right| = \frac{1}{k}\left|k - n - n(g_0 + g_a)\right| \tag{8}$$

But note that that $0 \leq g_0, g_a \leq \frac{k}{n} - 1$ implies $|k - n - n(g_0 + g_a)| \leq k - n$, so we can further bound (8) by:

$$\left|a - \frac{k_1}{k}d(m,n)\right| = 1 - \frac{n}{k} < 1 \tag{9}$$

Therefore, $\tilde{A}$ is maximally even. □

Lemma 3.3. *If $\tilde{A}$ is maximally even, then it has balanced gaps.*

Proof. Suppose that $\tilde{A}$ is maximally even, and let A be any unfolding of $\tilde{A}$. Then consider any substring of the form 10^{g_e} in A. Use $(*)$ on this substring to get the following cases: either $\lfloor \frac{n}{k}(g_e + 1) \rfloor = 1$ or $\lceil \frac{n}{k}(g_e + 1) \rceil = 1$. The first case is equivalent to $1 \leq \frac{n}{k}(g_e + 1) < 2$, which implies $\frac{k}{n} - 1 \leq g_e$. Then use $(*)$ on the substring 0^{g_e} to get $\lfloor \frac{n}{k} g_e \rfloor = 0$, which implies $g_e < \frac{k}{n}$. Combine these to get $\frac{k}{n} - 1 \leq g_e < \frac{k}{n}$, which is exactly $g_e = \lfloor \frac{k}{n} - 1 \rfloor$. The second case is equivalent to $0 < \frac{n}{k}(g_e + 1) \leq 1$, which implies $g_e \leq \frac{k}{n} - 1$. Then use $(*)$ on the substring $10^{g_e}1$ to get, in particular, $1 < \frac{n}{k}(g_e + 2)$, which implies $\frac{k}{n} - 2 < g_e$. Combine these to get $\frac{k}{n} - 2 < g_e \leq \frac{k}{n} - 1$, which is exactly $g_e = \lceil \frac{k}{n} - 1 \rceil$. □

This discussion of balanced gap necklaces leads us to the following construction, which should be seen as the key step in the development of this theory.

Definition 3.4. Let $\tilde{A}$ be a binary necklace with balanced gaps whose type is (k, n) with $n > 0$. If n does not divide k, then we can construct a new binary necklace by substituting

$$10^{\lfloor \frac{k}{n} - 1 \rfloor} \mapsto 0 \text{ and } 10^{\lceil \frac{k}{n} - 1 \rceil} \mapsto 1. \tag{10}$$

The resulting necklace is called the *compression* of $\tilde{A}$, denoted compress($\tilde{A}$). If it is the case that n divides k, then we use the convention that compress($\tilde{A}$) is the binary necklace of length n which consists only of 0's.

Remark 3.5. If $\tilde{A}$ is any binary necklace of type (k, n), then compress($\tilde{A}$) has type $(n, k \mod n)$. In particular, we see that the number of 1's in the compressed necklace is strictly less than the number of 1's in the original necklace.

It should be noted that, to be fully precise with this definition, the type of the binary necklace should parameterize a family of compress maps. This abuse of notation will be convenient in our mathematical notation, but it will be addressed more carefully in our algorithms. Algorithm 2 contains an implementation of the compress map, and it is clear that it runs in $\Theta(k)$ time and $\Theta(n)$ space.

Now we can proceed to the main lemma of this section:

Lemma 3.6. *A binary necklace $\tilde{A}$ with balanced gaps is maximally even if and only if compress($\tilde{A}$) is maximally even.*

Algorithm 2 Compressing a binary string

```
procedure COMPRESS(A, k, n)
    C ← LIST(n)
    G ← GETGAPS(A)
    for e ∈ {0, ... n − 1} do
        if G[e] = ⌊k/n − 1⌋ then
            C[e] ← 0
        else
            C[e] ← 1
    return C
```

Proof. Suppose that $\tilde{A}$ has balanced gaps and is of type (k, n). The claim is easy when n divides k, since the following must hold: $\tilde{A}$ is maximally even because of Lemma 3.2, and compress($\tilde{A}$) is maximally even because its type is $(n, 0)$. Now we proceed assuming that n does not divide k, which allow us to use the useful identity:

$$\left\lceil \frac{k}{n} - 1 \right\rceil - \left\lfloor \frac{k}{n} - 1 \right\rfloor = 1 \tag{11}$$

First suppose that $\tilde{A}$ of type (k, n) with balanced gaps is not maximally even. Then by Lemma 2.3, there exists some unfolding A of $\tilde{A}$ and some indices $i, j \in \mathbb{Z}/k\mathbb{Z}$ that satisfy

$$\#(A \cap [i, j)_k) \geq \frac{n}{k} \vec{d}_k(i, j) + 1. \tag{12}$$

Let us assume that $i, j \in \mathbb{Z}/k\mathbb{Z}$ are such that they minimize the value of $\vec{d}_k(i, j)$. Then taking $a = \#(A \cap [i, j)_k)$, we can write $A \cap [i, j)_k = 10^{g_1}1 \ldots 10^{g_a}1$ with $g_e = \lfloor \frac{k}{n} - 1 \rfloor$ or $\lceil \frac{k}{n} - 1 \rceil$ for all $e = 1, \ldots a$. Then (12) becomes

$$a + 1 \geq \frac{n}{k}\left(a + 1 + \sum_{e=1}^{a} g_e\right) + 1. \tag{13}$$

Letting r denote the number of $e \in \{1, \ldots a\}$ for which $g_e = \lceil \frac{k}{n} - 1 \rceil$, we see that (13) can be expressed as

$$\begin{aligned} a + 1 &\geq \frac{n}{k}\left(a + 1 + (a - r)\left\lfloor \frac{k}{n} - 1 \right\rfloor + r\left\lceil \frac{k}{n} - 1 \right\rceil\right) + 1 \\ &= \frac{n}{k}\left(a + 1 + a\left\lfloor \frac{k}{n} - 1 \right\rfloor + r\right) + 1, \end{aligned} \tag{14}$$

where the equality follows from identity (11). Then check that (14) rearranges to

$$r \leq \left(\frac{k}{n} - 1 - \left\lfloor \frac{k}{n} - 1 \right\rfloor\right)a - 1 = \left(\frac{k \mod n}{n}\right)a - 1. \tag{15}$$

But observe that, by definition, r is exactly the number of 1's in the image of $A \cap [i, j)_k$ under the compress map. Therefore, compress($\tilde{A}$) is not maximally even.

For the second direction, suppose that $\tilde{A}$ of type (k,n) with balanced gaps is such that $\tilde{C} = \text{compress}(\tilde{A})$ is not maximally even. Then there must exist an unfolding C of $\tilde{C}$ and indices $i,j \in \mathbb{Z}/n\mathbb{Z}$ such that, for $r = \#(C \cap [i,j)_k)$ and $a = \vec{d}_k(i,j)$, we get

$$r \leq \left(\frac{k \mod n}{n}\right)a - 1 = \left(\frac{k}{n} - 1 - \left\lfloor \frac{k}{n} - 1 \right\rfloor\right)a - 1. \tag{16}$$

But observe that (16) is equivalent to (15), so it rearranges to (14). Then note that there exists an unfolding A of $\tilde{A}$ and indices $i', j' \in \mathbb{Z}/k\mathbb{Z}$ such that $C \cap [i,j)_k$ is the image of $A \cap [i',j')_k = 10^{g_1} \dots 10^{g_a}$ under the compress map. This implies $a + (a-r)\lfloor \frac{k}{n} - 1 \rfloor + r\lceil \frac{k}{n} - 1 \rceil = \vec{d}_k(i,j)$. Then we have

$$a + 1 \geq \frac{n}{k}(\vec{d}_k(i',j') + 1) + 1. \tag{17}$$

Let us consider this inequality with respect to the string $A \cap [i', j'+1) = 10^{g_1} \dots 10^{g_a} 1$. On the left side, $a+1$ is equal to $\#(A \cap [i', j'+1))$, and, on the right side $\vec{d}_k(i',j')+1$, is equal to the length of the interval in question. That is, (17) implies that $\tilde{A}$ is not maximally even, which completes the proof. □

Theorem 3.7. *A binary necklace $\tilde{A}$ is maximally even if and only if compress$^m(\tilde{A})$ has type $(k', 0)$ for some nonnegative integer m and positive integer k'.*

Proof. First, let $\tilde{A}$ be a maximally even binary necklace. Then use Proposition 3.6 inductively to see that every necklace that is an element of the sequence $\{\text{compress}^0(\tilde{A}), \text{compress}^1(\tilde{A}), \text{compress}^2(\tilde{A}), \dots\}$ is maximally even. Denoting the respective types of these necklaces by $\{(k_0,n_0),(k_1,n_1),((k_2,n_2),\dots\}$, Remark 3.5 guarantees that we must have $n_0 > n_1 > n_2 > \dots$. As this is a strictly decreasing sequence of positive integers, there must be some nonnegative integer r such that $n_m = 0$ holds. Then we see that compress$^m(\tilde{A})$ has type $(k_m, 0)$, as desired.

Conversely, suppose that $\tilde{A}$ of type (k,n) is such that compress$^m(\tilde{A})$ is of type $(k', 0)$ for some nonnegative integer m and some positive integer k'. Use Proposition 3.6 inductively to see that $\{\text{compress}^m(\tilde{A}), \text{compress}^{m-1}(\tilde{A}), \dots \text{compress}^0(\tilde{A})\}$ is a finite sequence of maximally even necklaces. In particular, we see that the element compress$^0(\tilde{A}) = \tilde{A}$ is maximally even, as desired. □

We would be remiss not to state and prove a uniqueness result for maximal evenness in this paper. While such a result is not novel by any means (see Section 4, Proposition 5A of [7]), the theory developed in this section reveals an elegant proof of the well-known fact:

Theorem 3.8. *There is a unique maximally even binary necklace for each type. That is, if A_1 and A_2 are maximally even binary strings of the same type, then they are translation-equivalent: $A_1 \sim A_2$.*

Proof. Let $\tilde{A}_1$ and $\tilde{A}_2$ be maximally even binary necklaces of the same type (k,n). By Theorem 3.7, there exist nonnegative integers m_1, m_2 and positive integers k_1', k_2' such that compress$^{m_1}(\tilde{A}_1)$ is of type $(k_1', 0)$ and compress$^{m_2}(\tilde{A}_2)$ is of type $(k_2', 0)$. However, Remark 3.5 guarantees that $m_1 = m_2$ and $k_1' = k_2'$. Then observe that every compress map is invertible (its inverse can be explicitly constructed if the

type of the domain necklace is known) hence injective. Since the composition of injective functions is injective, we conclude $\tilde{A}_1 = \tilde{A}_2$. □

Finally, we are able to complete the task of computationally checking whether a given binary necklace is maximally even; Theorem 3.7 suggests a recursive algorithm for doing this. While it is slow in its current form, its runtime can be improved dramatically by making the following modification: If necessary, flip the bits of compress($\tilde{A}$) in order to maintain that $n \leq \frac{k}{2}$ holds at every function call. In particular, types will be mapped under compress as follows: If $k \mod n < \frac{n}{2}$, then $(k, n) \mapsto (n, n - (k \mod n))$; if $k \mod n \geq \frac{n}{2}$, then $(k, n) \mapsto (n, n - (k \mod n))$. We record this procedure precisely in Algorithm 3.

Algorithm 3 Checking if a binary string is maximally even, recursively

```
1: procedure IsMaxEven(A)
2:     k ← length(A)
3:     n ← sum(A)
4:     return IsMaxEvenHelper(A, k, n)
5:
6: procedure IsMaxEvenHelper(A, k, n)
7:     if n = 0 then
8:         return true
9:     else if not HasBalancedGaps(A, k, n) then
10:        return false
11:    else if n divides k then
12:        return true
13:    else
14:        C ← Compress(A, k, n)
15:        if k mod n < n/2 then
16:            return IsMaxEvenHelper(C, n, k mod n)
17:        else
18:            C ← C ⊕ 1^n                                  ▷ Flip all of the bits
19:            return IsMaxEvenHelper(C, n, n − (k mod n))
```

Theorem 3.9. *For a binary string A of type (k, n), Algorithm 3 returns whether or not A is maximally even. It runs in $\Theta(k)$ time and in $\Theta(n)$ space. The depth of its function call stack is $O(\log_2 n)$.*

Proof. Theorem 3.7 proves the correctness of the unmodified algorithm. To see that the modification does not alter the correctness, simply recall that A is maximally even if and only if $A \oplus 1^k$ is maximally even.

To find the time complexity, recall that line 14 runs in $\Theta(k)$ time, and that all other lines run in $O(1)$ time. Since the modification guarantees that k drops by a factor of at least 2 at every function call, we see that the total time complexity is $\Theta(k) + \Theta(\frac{k}{2}) + \Theta(\frac{k}{4}) + \cdots = \Theta(k)$. Likewise, line 14 runs in $\Theta(n)$ space and all other lines are $O(1)$ space. Since the modification guarantees that n also drops by a factor of at least 2 at every function call, we see that the total space complexity is $\Theta(n) + \Theta(\frac{n}{2}) + \Theta(\frac{n}{4}) + \cdots = \Theta(n)$. The exponential decay of n also guarantees that the function call stack has a depth of $O(\log_2 n)$. □

4. Generalizations

In this section, we consider two natural generalizations of maximal evenness using a discrepancy characterization. Then, we pose some open questions about these generalized scenarios. By design, the mathematical work in this section is rather informal and meant only to inspire new ideas and directions.

First, we consider what it means to be maximally even in a torus of dimension greater than one. To do this, let $k_1, \dots k_d$ be positive ingers, and let us embed $\mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_d\mathbb{Z}$ inside $(\mathbb{S}^1)^d$ in the obvious way. Then for any subset $A \subseteq \mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_d\mathbb{Z}$ with $|A| = n$, we let its *type* be $(k_1, \dots k_d, n)$. This allos us to define:

Definition 4.1. A set A of type $(k_1, \dots k_d, n)$ is called *maximally even* if it satisfies the discrepancy bound

$$\sup_{I \subseteq (\mathbb{S}^1)^d} \left| \frac{\#(A \cap I)}{n} - \lambda(I) \right| < \frac{1}{n} + E(k_1, \dots k_d) \tag{18}$$

where the supremum is taken over all rectangles of $(\mathbb{S}^1)^d$, and E is the function

$$E(k_1, \dots k_d) = 1 - \prod_{i=1}^{d} \left(1 - \frac{1}{k_i}\right). \tag{19}$$

While this function E may seem mysterious at first, we remark that it in fact represents the error incurred by "snapping" a rectangle $I \subseteq (\mathbb{S}^1)^d$ to the grid $\mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_d\mathbb{Z}$.

Question 4.2. *Which sets of type $(k_1, \dots k_d, n)$ are maximally even? Does there exist a unique maximally even set for each type, up to translation-equivalence? What is a fast algorithm for determining whether a given set is maximally even?*

Next, we consider what it means to be maximally even in the integers. To do this, notice that the term $\frac{n}{k}\vec{d}_k(i,j)$ may be interpreted as the product of a density and a length. Hence, we may define maximal evenness for subsets of $\mathbb{Z}$ as follows:

Definition 4.3. Say that $A \subseteq \mathbb{Z}$ *maximally even* if there exists some real number $\alpha \in [0,1]$ such that the following discrepancy bound is satisfied:

$$\sup_{I \subseteq \mathbb{R}} \left| \#(A \cap I) - \alpha\lambda(I) \right| < 1, \tag{20}$$

where λ is the Lebesgue measure on $\mathbb{R}$, and the supremum is taken over all intervals (connected subsets) of $\mathbb{R}$. In this case, α is called the *density* of A.

We can justify the use of the name "density" given to α by proving that α is in fact the natural density of A. However, the converse is not true general. While natural density is only required to hold asymptotically, Definition 4.3 requires that a density condition holds on every interval of $\mathbb{R}$. As such, density in the sense of Definition 4.3 is an extremely strong requirement.

Lemma 4.4. *Suppose that $A \subseteq \mathbb{Z}$ is maximally even with density α. Then the natural density of A is α.*

Proof. Suppose that $A \subseteq \mathbb{Z}$ is maximally even with density α. Then consider the sequence of intervals $I_n = [-n, n]$, which have length $\lambda(I_n) = 2n + 1$. It follows from maximal evenness that we have

$$\left| \frac{\#(A \cap [-n, n])}{2n+1} - \alpha \right| < \frac{1}{2n+1}. \tag{21}$$

Taking the limit as $n \to \infty$ gives

$$\lim_{n \to \infty} \frac{\#(A \cap [-n, n])}{2n+1} = \alpha, \tag{22}$$

which is exactly the desired result. □

Question 4.5. *Which sequences $A \subseteq \mathbb{Z}$ of a given density α are maximally even? Does there exist a unique sequence, up to translation-equivalence?*

Lastly, we ask how to generalize all of these notions of maximal evenness within a common framework. How can we capture both $\mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_d\mathbb{Z} \subseteq (\mathbb{S}^1)^d$ and $\mathbb{Z} \subseteq \mathbb{R}$? How should we define maximal evenness for subsets $A \subseteq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$?

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